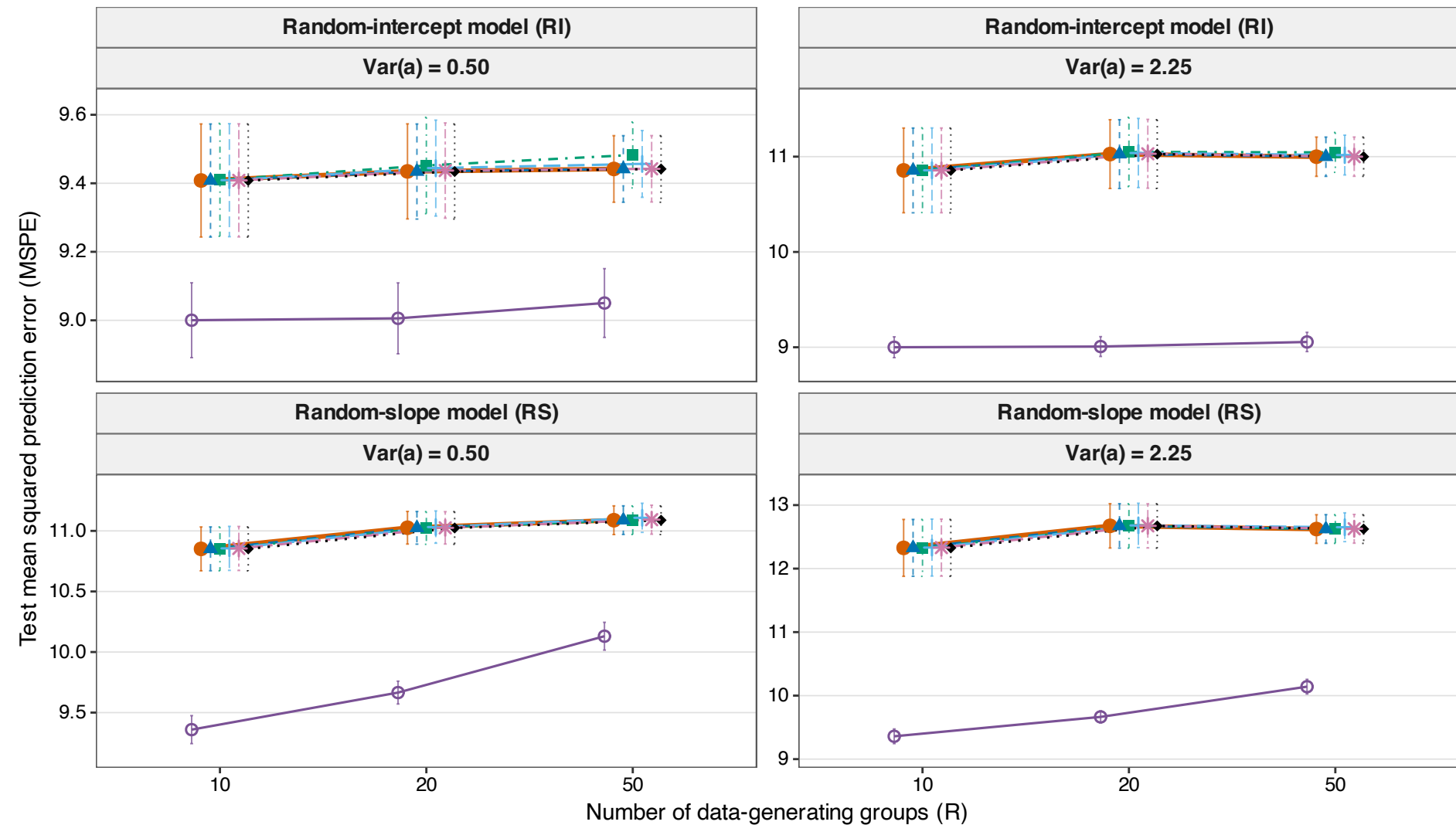


Case 1: Independent uniform covariates

Mean test MSPE with 95% Monte Carlo error bars; each panel has an independent y-scale

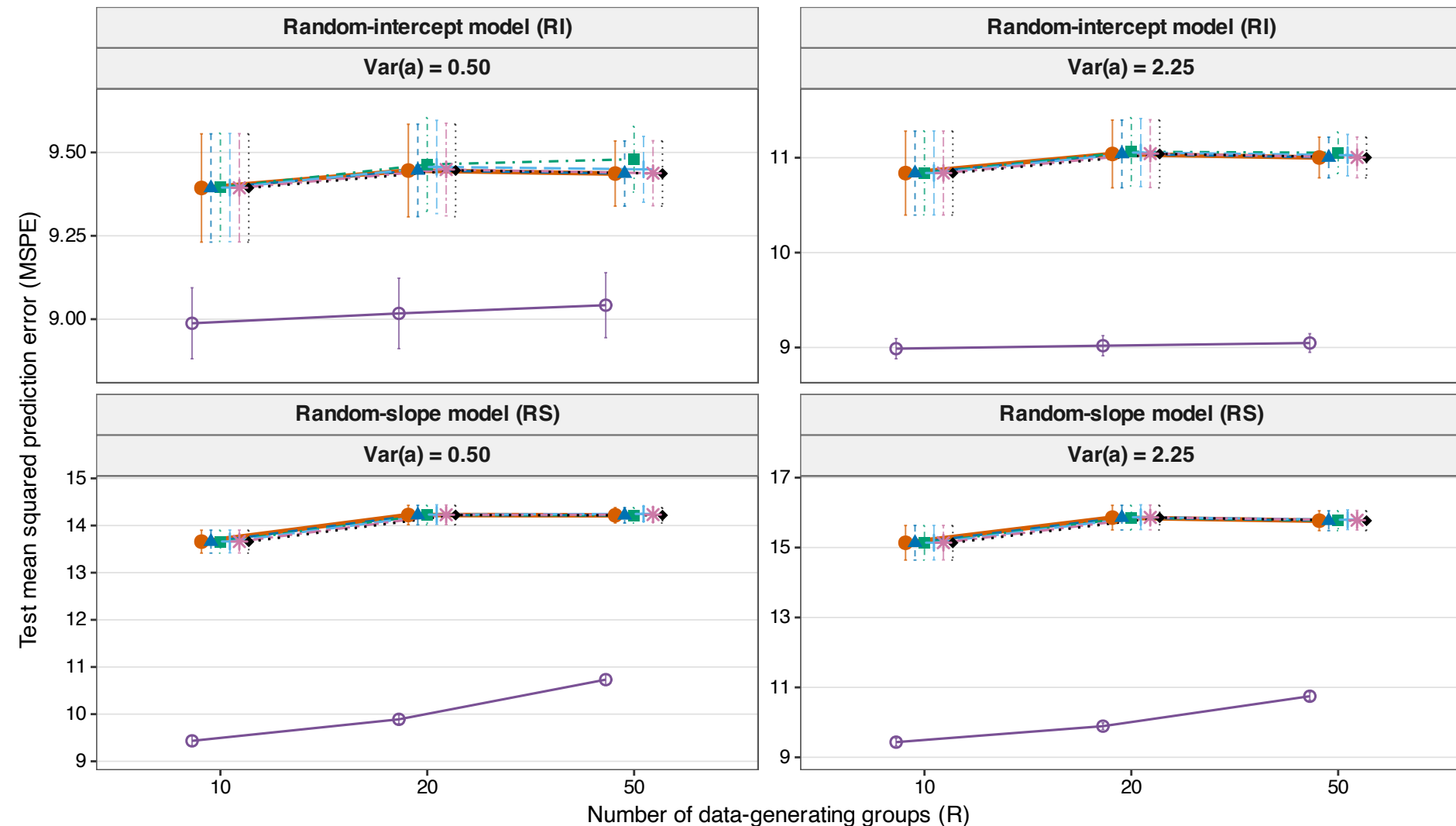


ORACLE CIRG KM BLM CPF OBS LM

Oracle uses the true generating groups and is an infeasible benchmark; all other methods use no true-group labels.

Case 2: Correlated Gaussian covariates

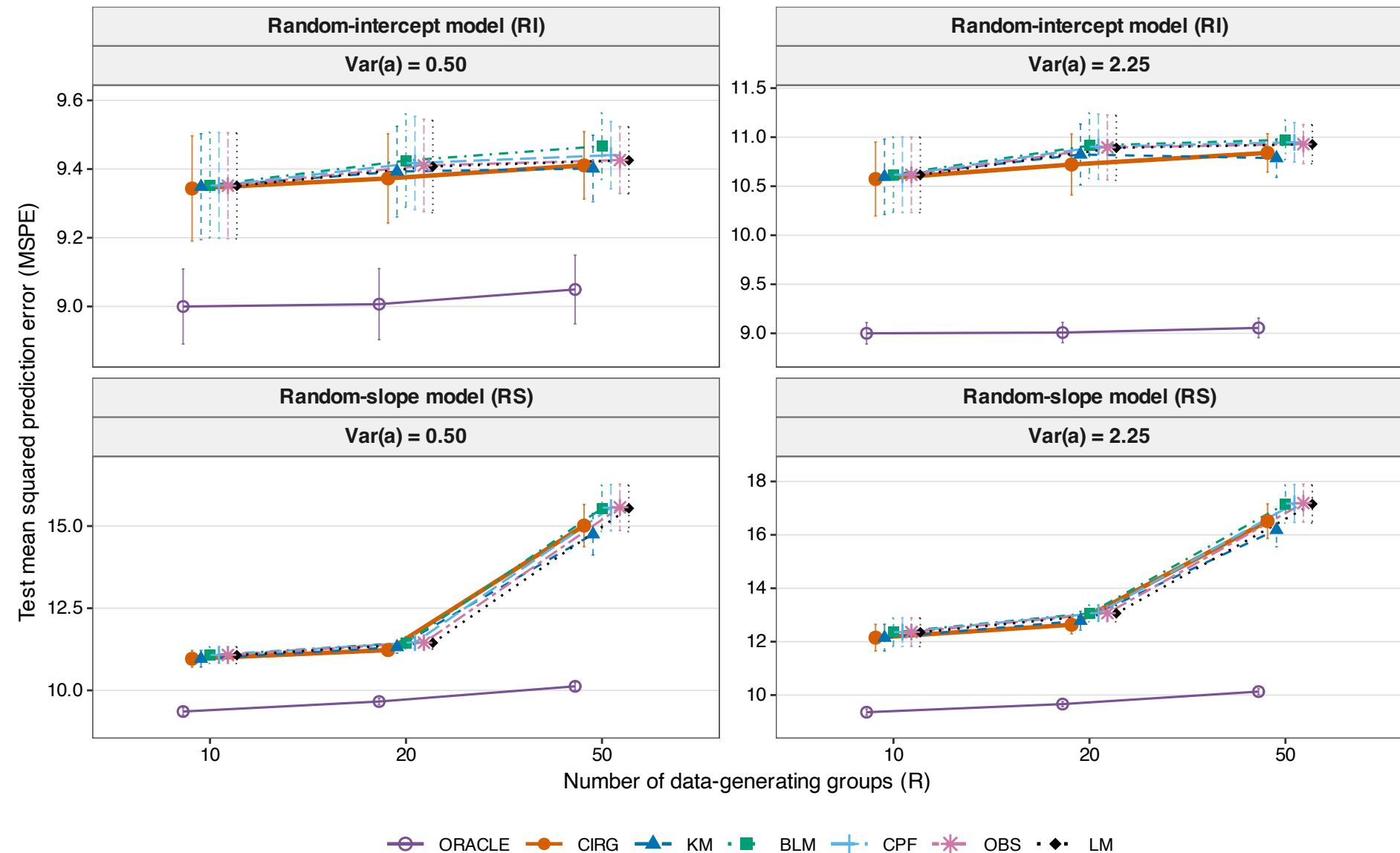
Mean test MSPE with 95% Monte Carlo error bars; each panel has an independent y-scale



Oracle uses the true generating groups and is an infeasible benchmark; all other methods use no true-group labels.

Case 3: Group-shifted uniform covariates

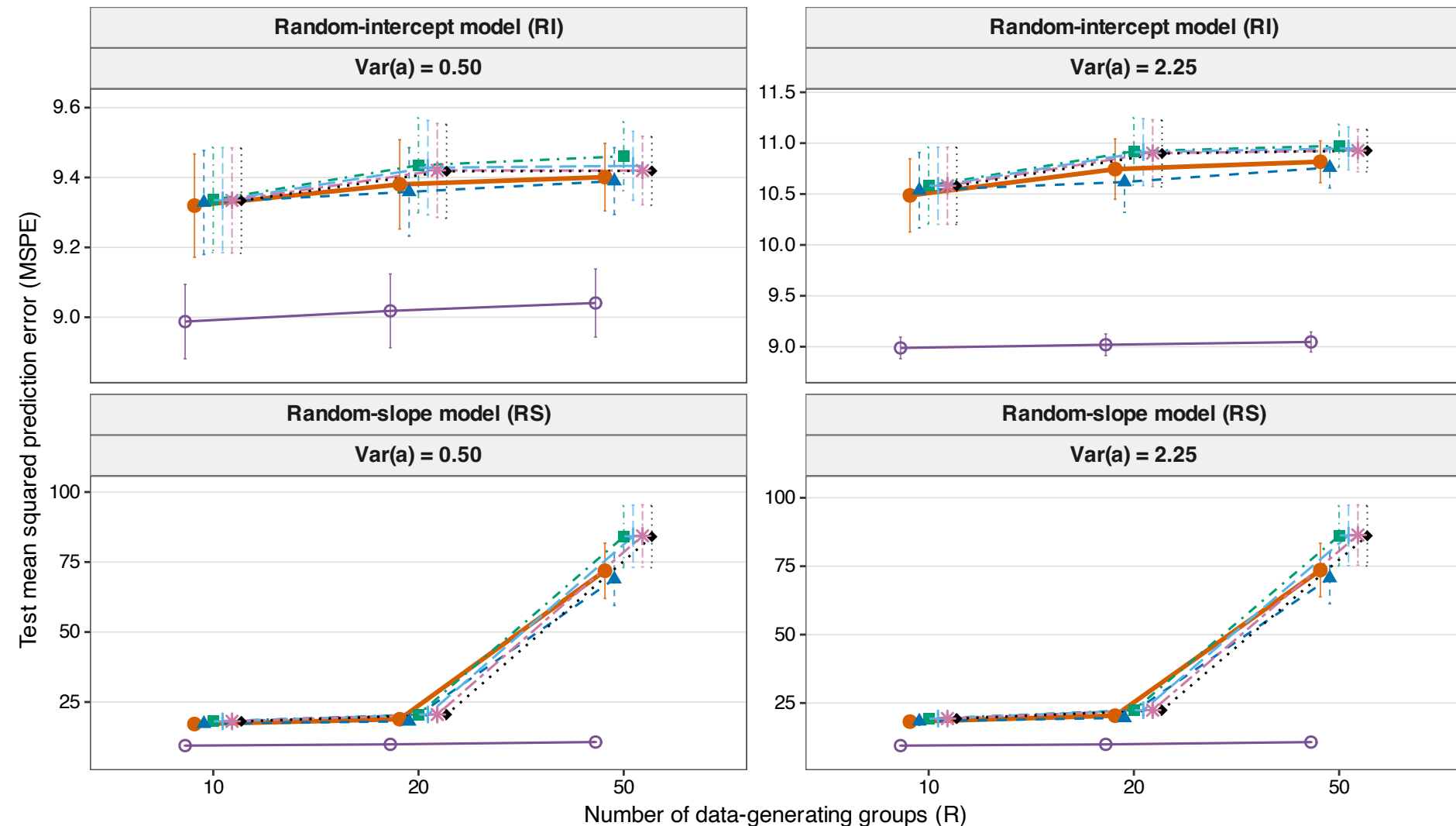
Mean test MSPE with 95% Monte Carlo error bars; each panel has an independent y-scale



Oracle uses the true generating groups and is an infeasible benchmark; all other methods use no true-group labels.

Case 4: Group-shifted Gaussian covariates

Mean test MSPE with 95% Monte Carlo error bars; each panel has an independent y-scale

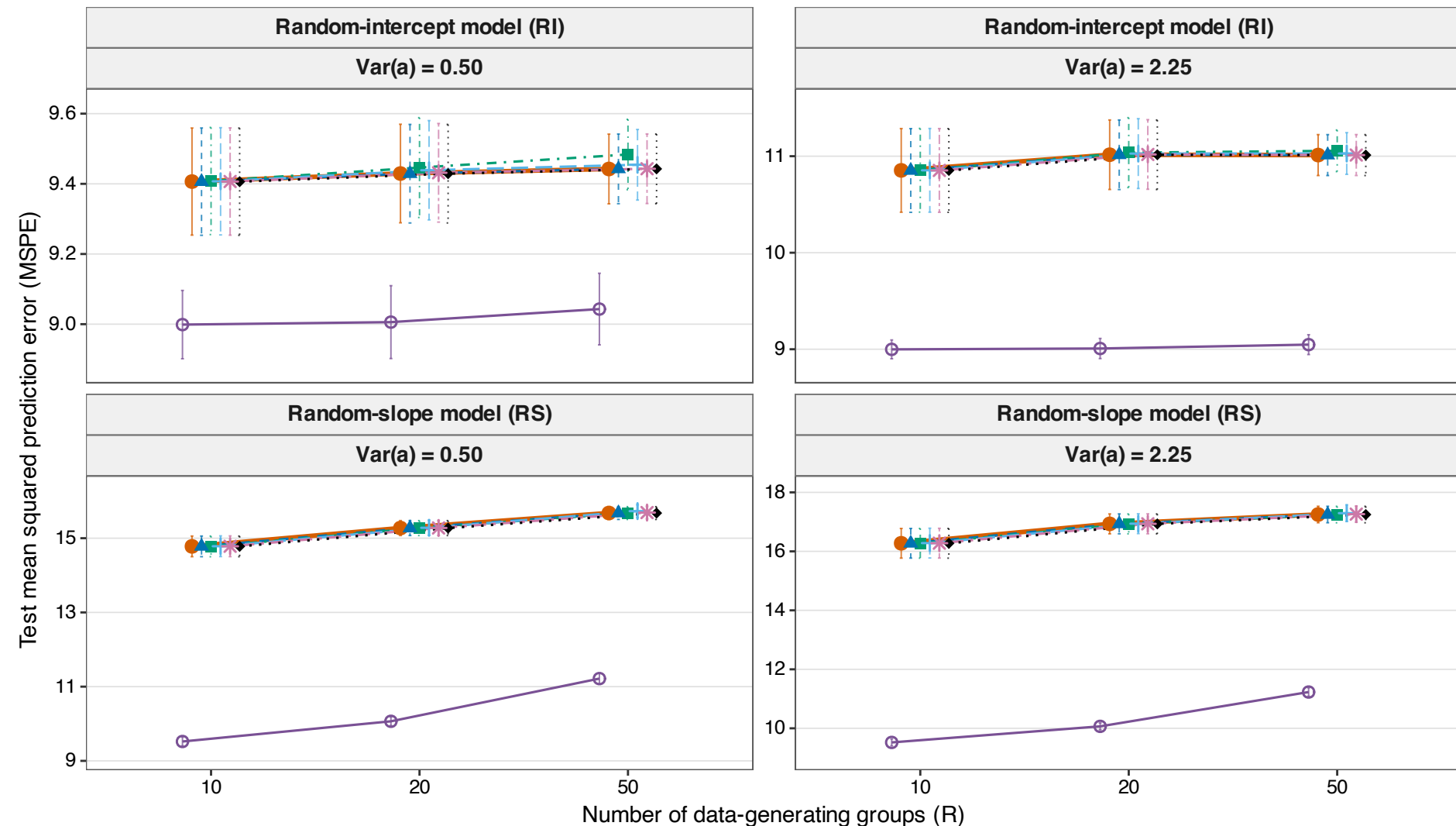


—○— ORACLE —●— CIRG —▲— KM —■— BLM —+— CPF —*— OBS —◆— LM

Oracle uses the true generating groups and is an infeasible benchmark; all other methods use no true-group labels.

Case 5: Heavy-tailed t covariates

Mean test MSPE with 95% Monte Carlo error bars; each panel has an independent y-scale

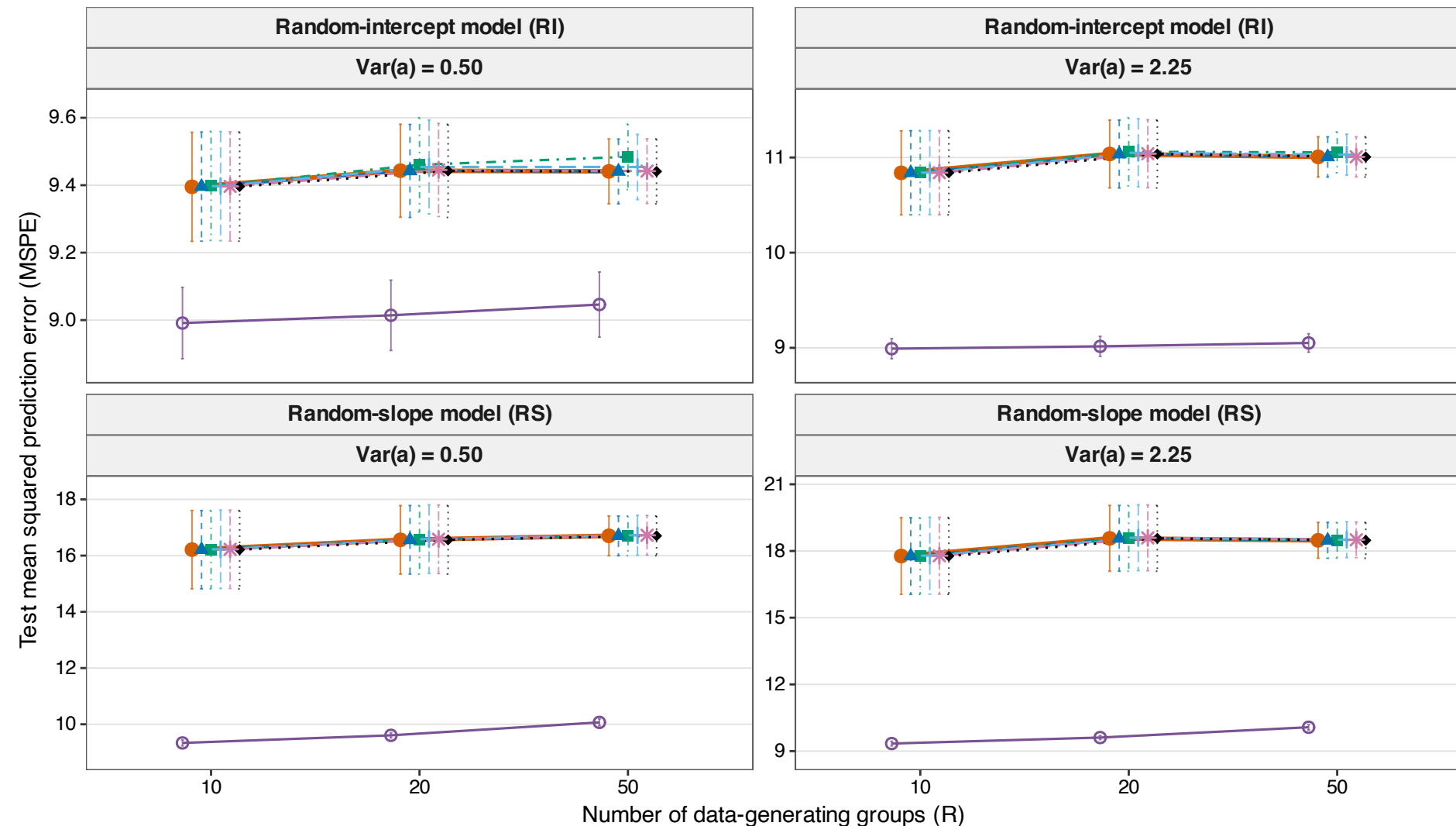


ORACLE CIRG KM BLM CPF OBS LM

Oracle uses the true generating groups and is an infeasible benchmark; all other methods use no true-group labels.

Case 6: Log-normal covariates

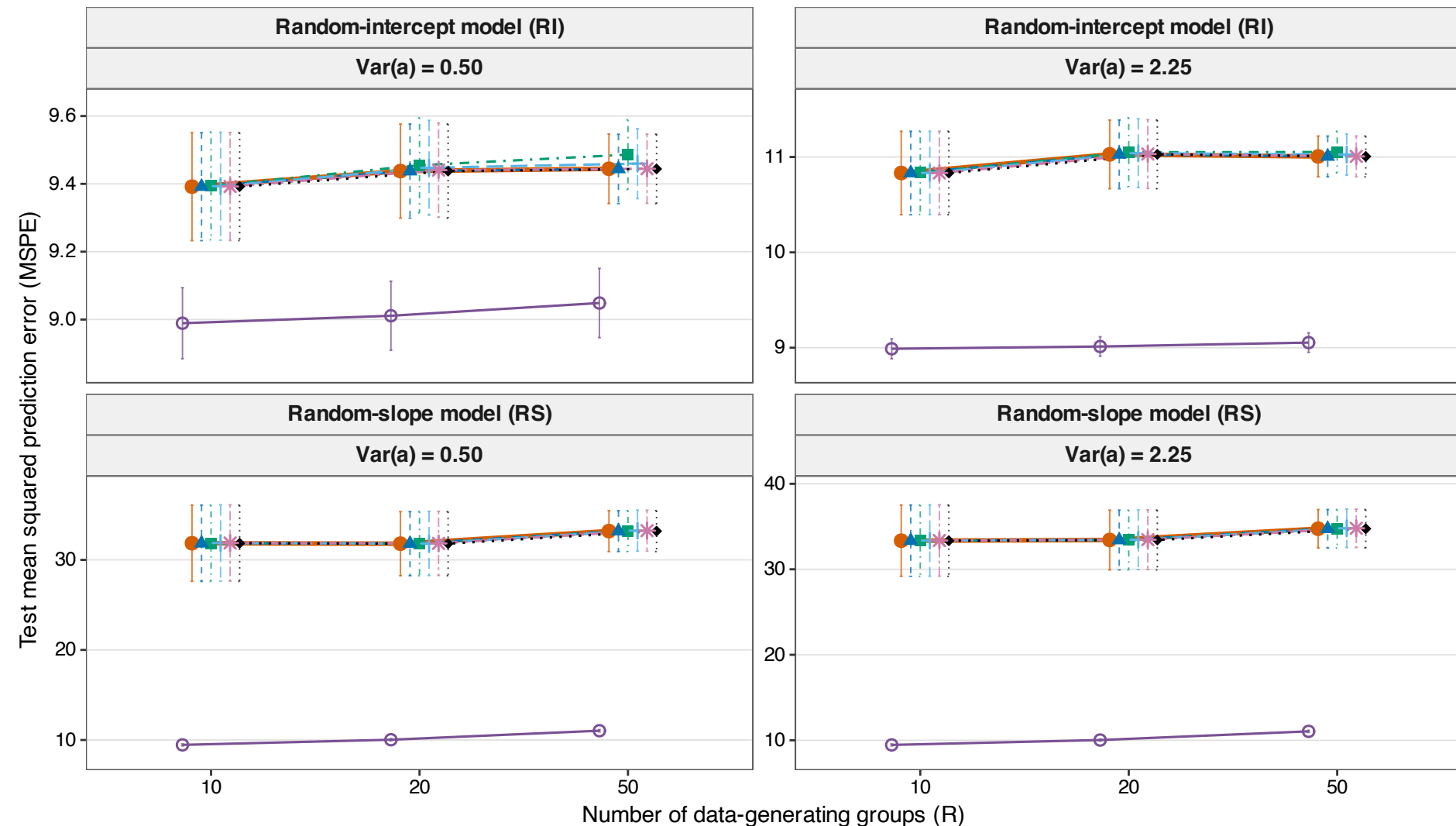
Mean test MSPE with 95% Monte Carlo error bars; each panel has an independent y-scale



Oracle uses the true generating groups and is an infeasible benchmark; all other methods use no true-group labels.

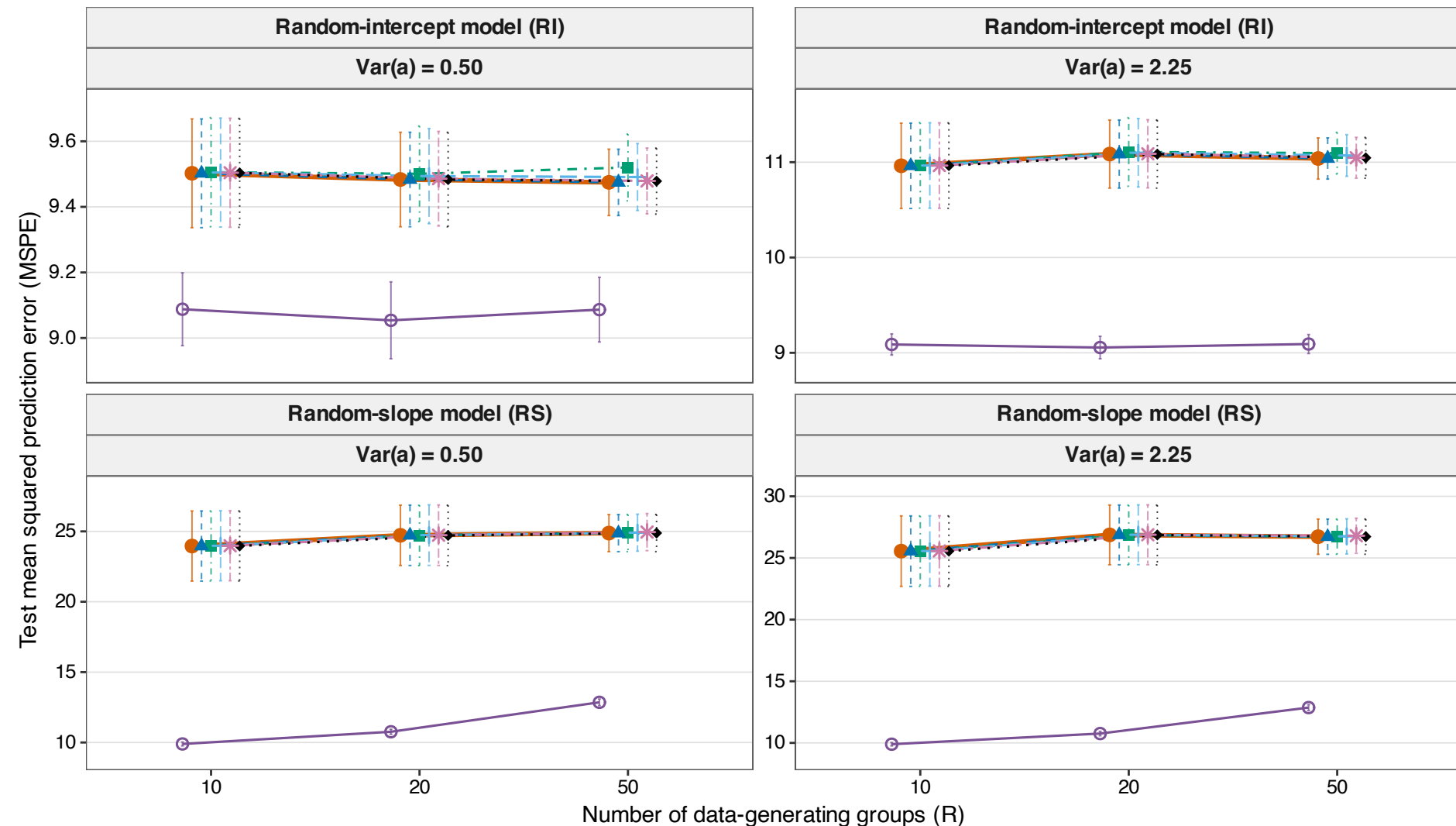
Case 7: Two-component Gaussian mixture

Mean test MSPE with 95% Monte Carlo error bars; each panel has an independent y-scale



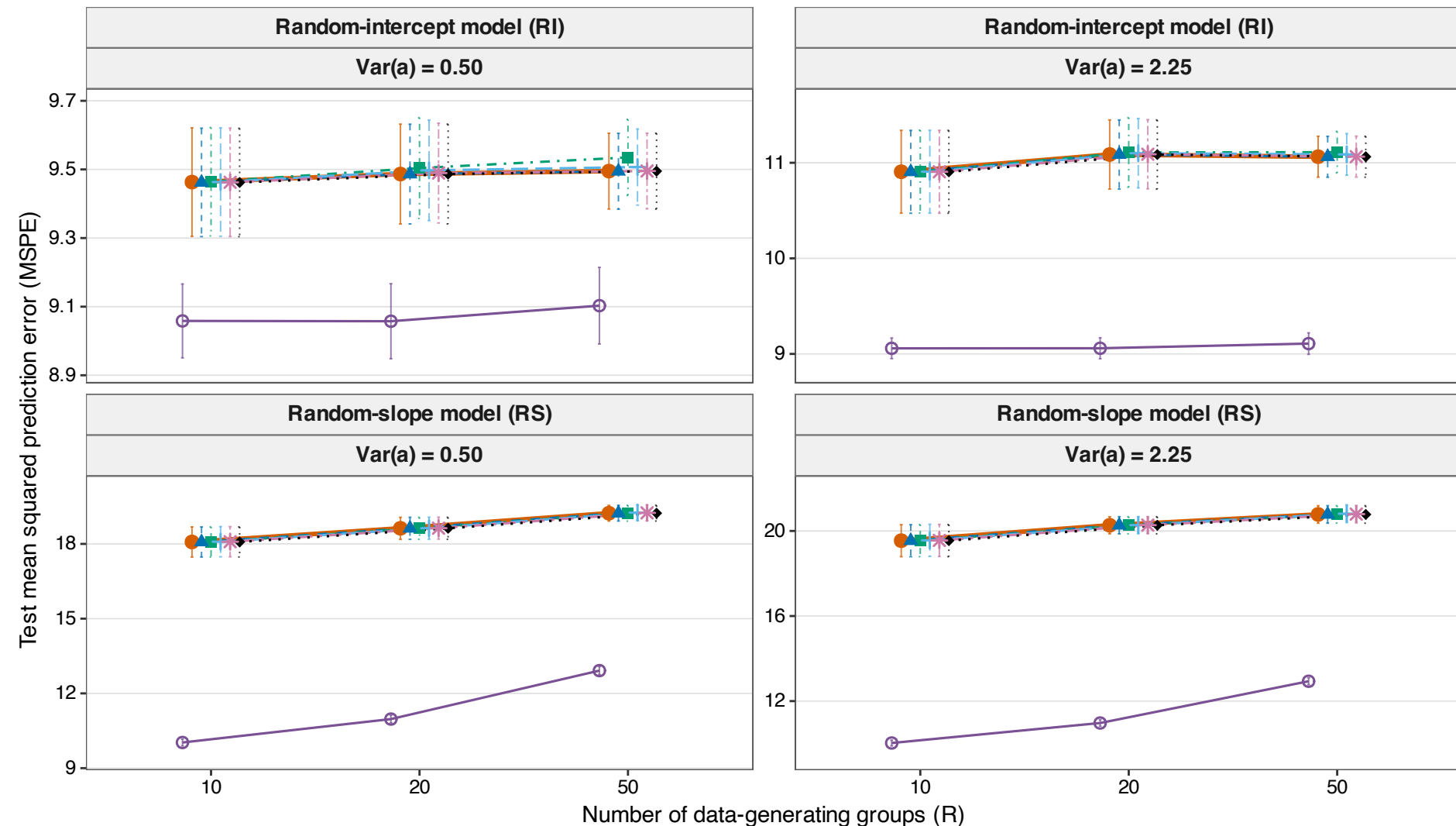
Case 8: Mean covariate shift

Mean test MSPE with 95% Monte Carlo error bars; each panel has an independent y-scale



Case 9: Covariance shift

Mean test MSPE with 95% Monte Carlo error bars; each panel has an independent y-scale

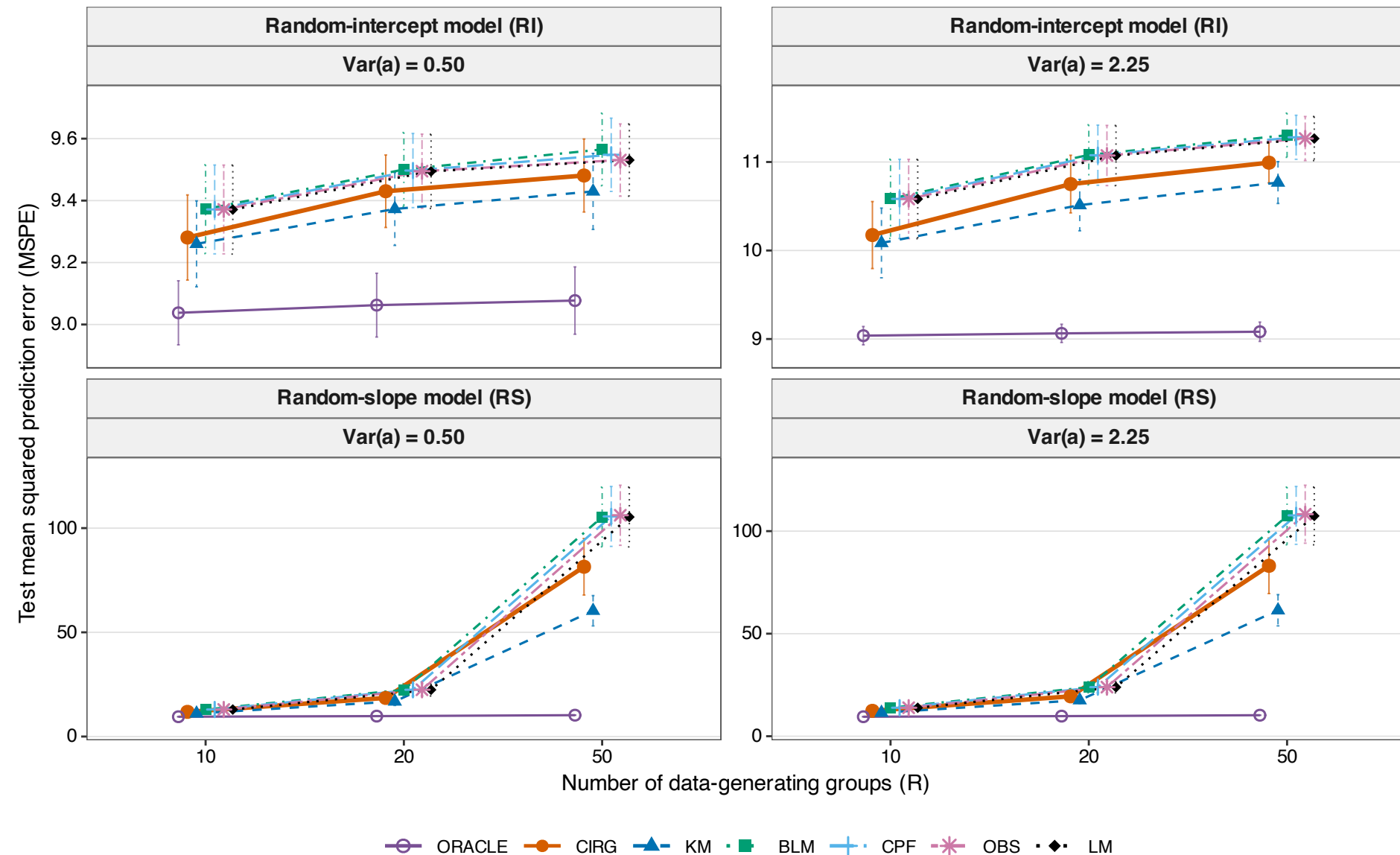


ORACLE CIRG KM BLM CPF OBS LM

Oracle uses the true generating groups and is an infeasible benchmark; all other methods use no true-group labels.

Case 10: Type structure with nuisance modes

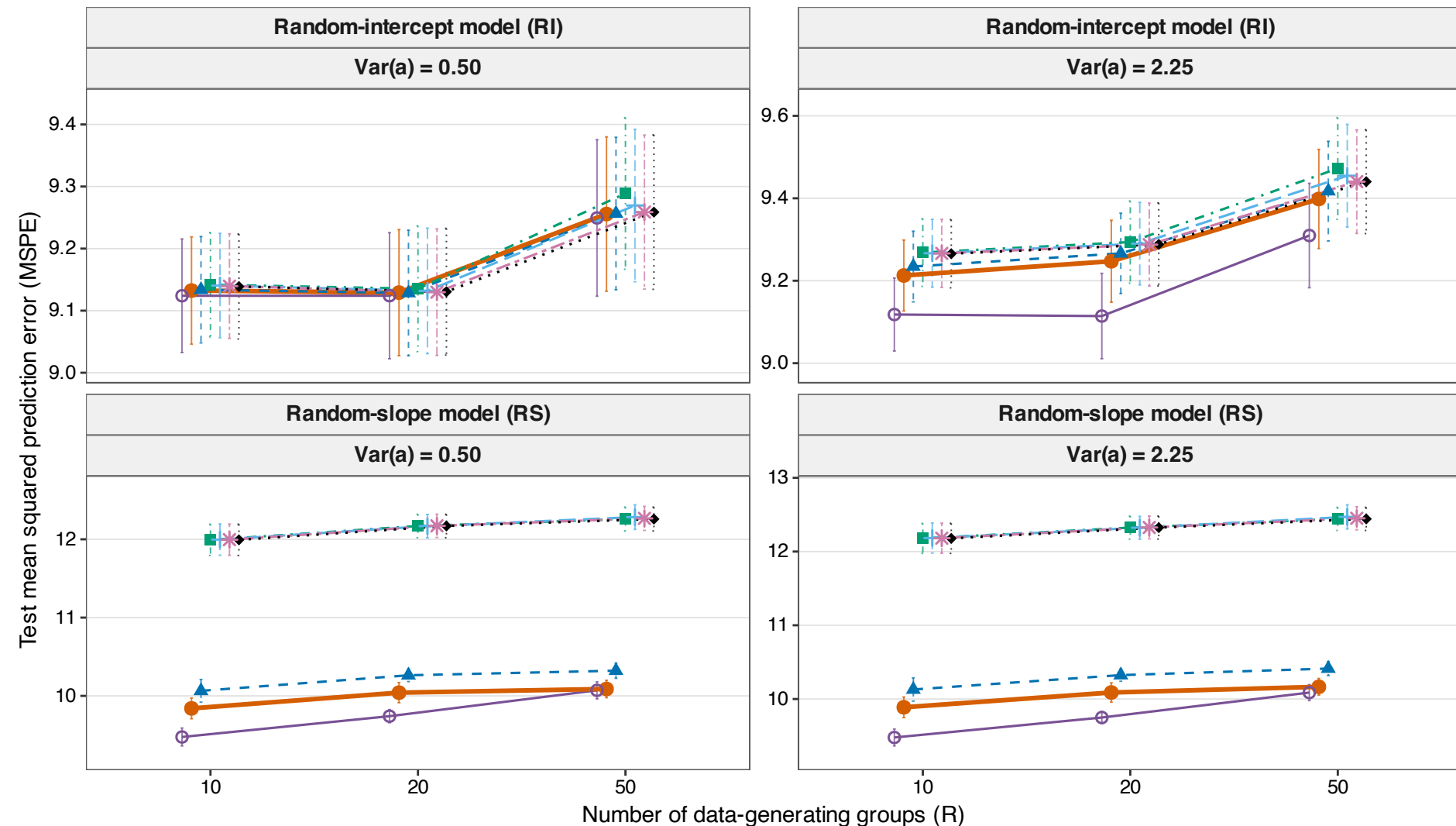
Mean test MSPE with 95% Monte Carlo error bars; each panel has an independent y-scale



Oracle uses the true generating groups and is an infeasible benchmark; all other methods use no true-group labels.

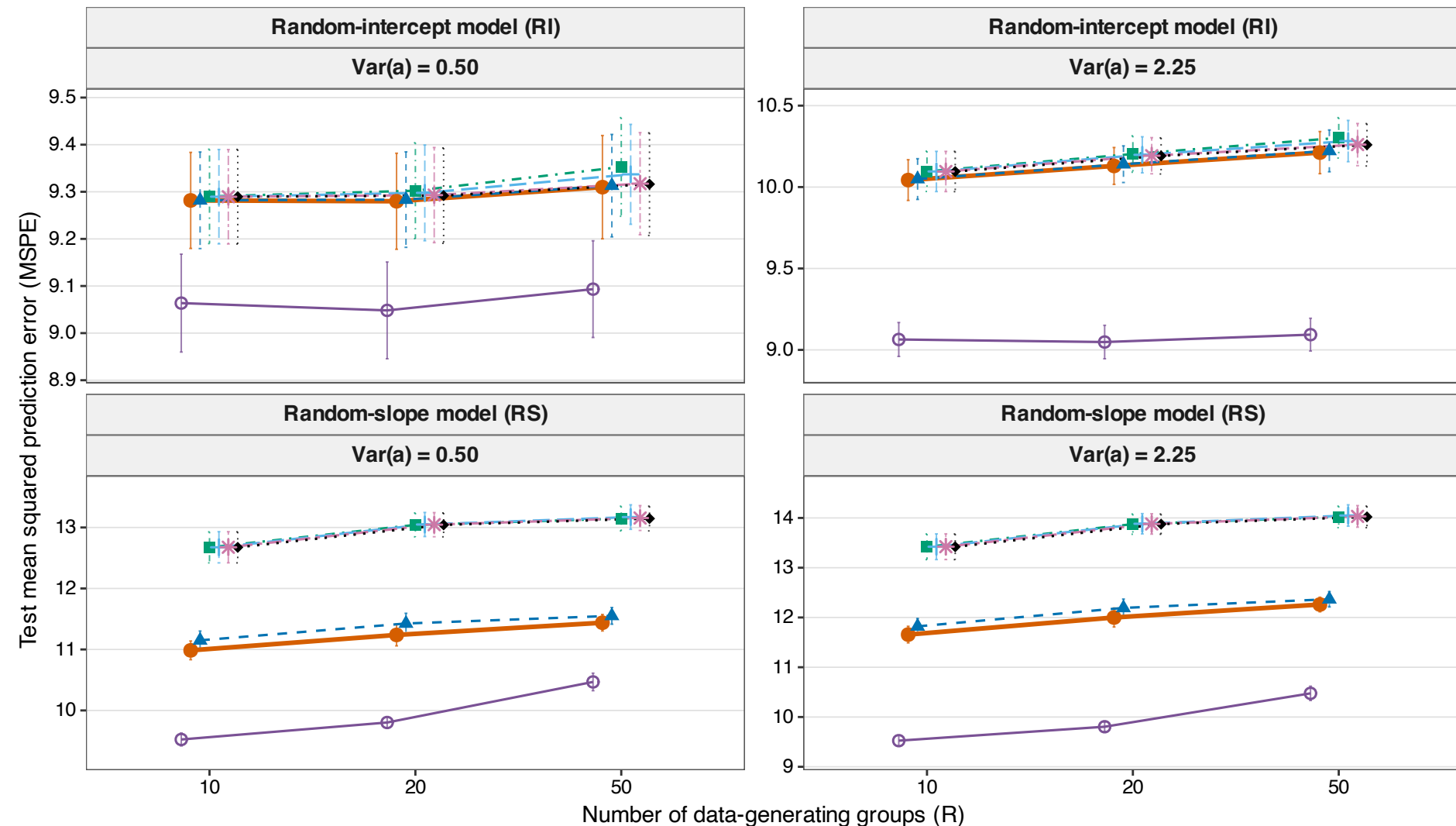
Case 11: Structured latent predictive effects

Mean test MSPE with 95% Monte Carlo error bars; each panel has an independent y-scale



Case 12: Continuous predictive heterogeneity

Mean test MSPE with 95% Monte Carlo error bars; each panel has an independent y-scale

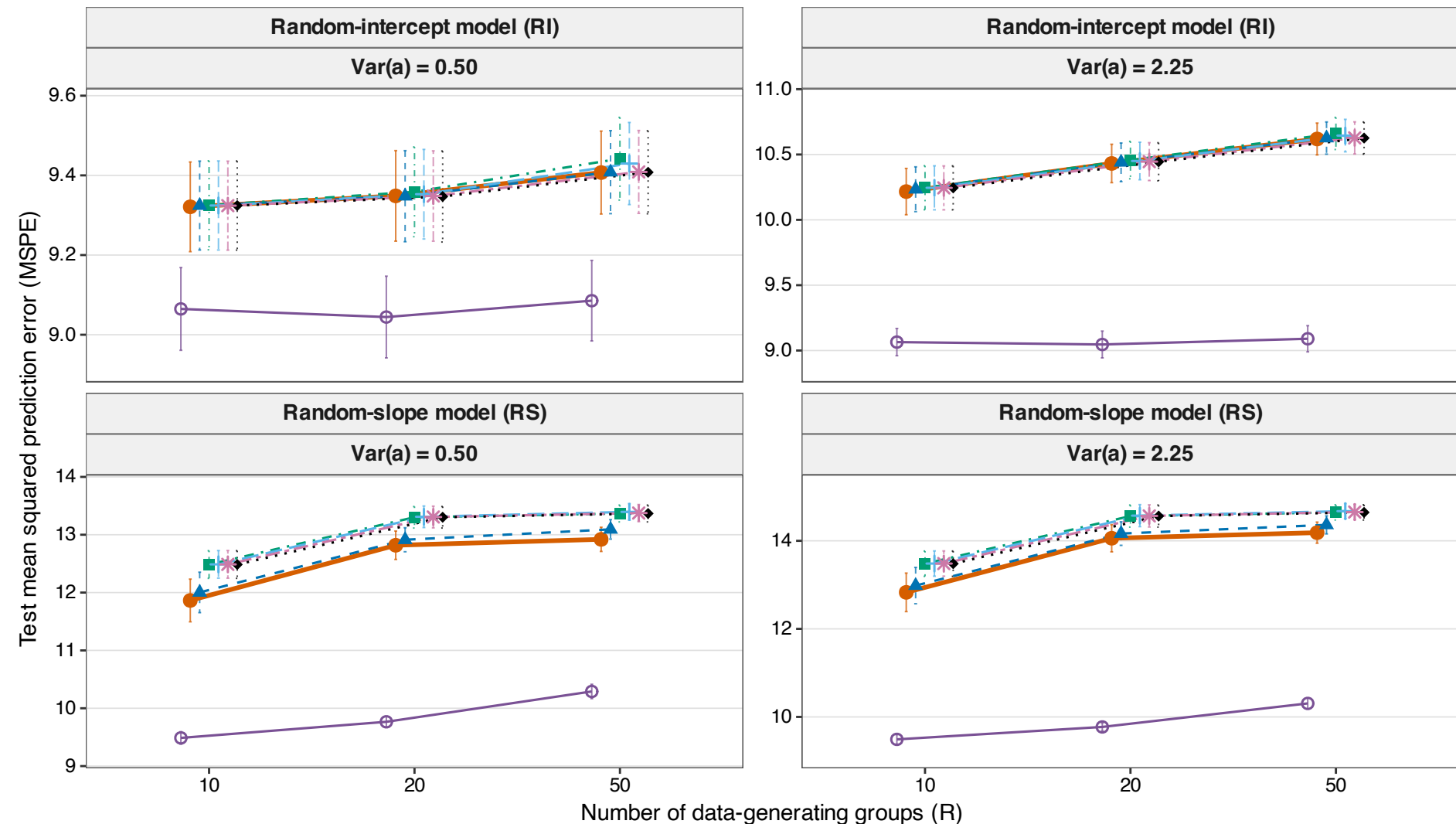


ORACLE CIRG KM BLM CPF OBS LM

Oracle uses the true generating groups and is an infeasible benchmark; all other methods use no true-group labels.

Case 13: Density-prediction conflict

Mean test MSPE with 95% Monte Carlo error bars; each panel has an independent y-scale

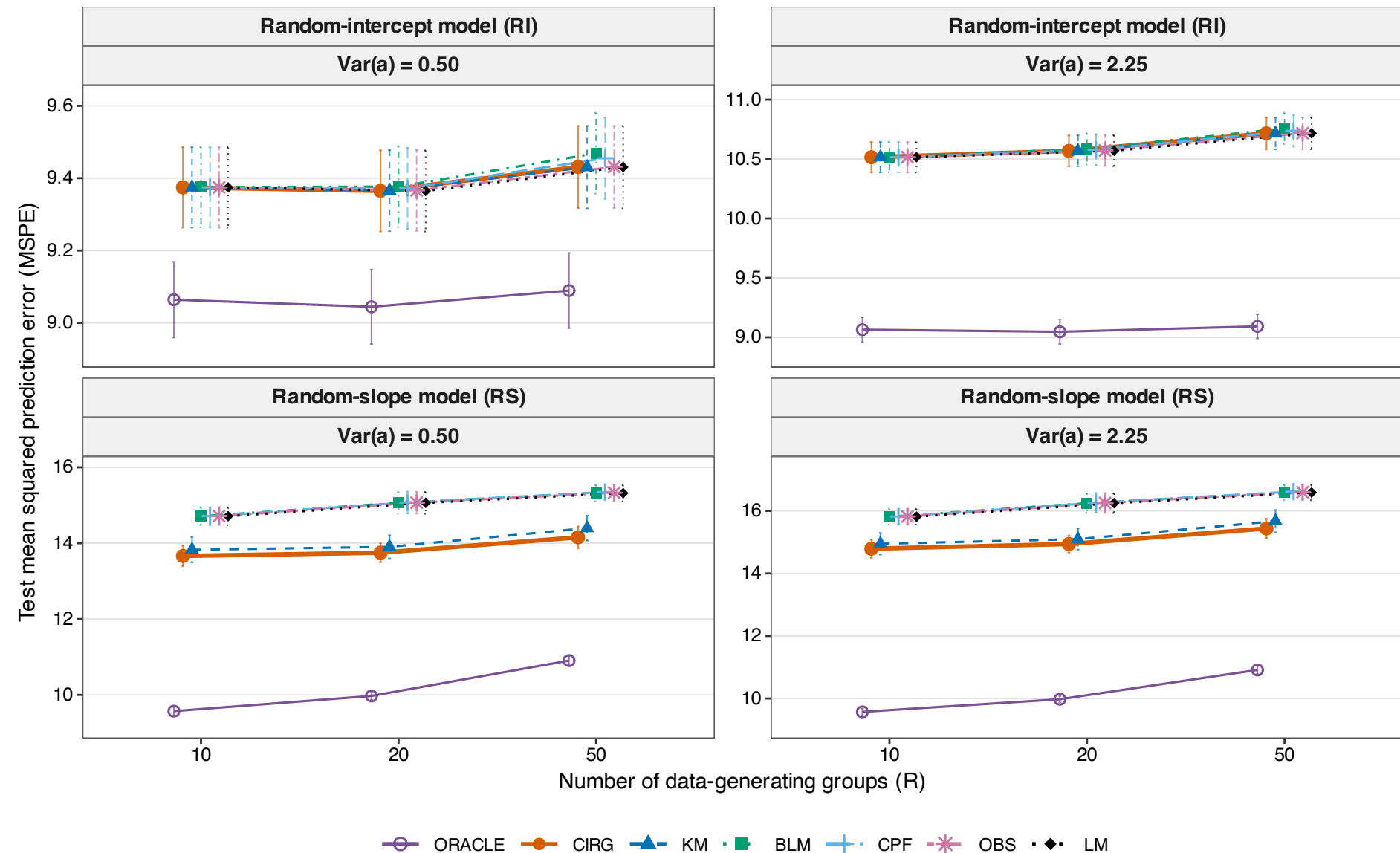


ORACLE CIRG KM BLM CPF OBS LM

Oracle uses the true generating groups and is an infeasible benchmark; all other methods use no true-group labels.

Case 14: Overlapping predictive states

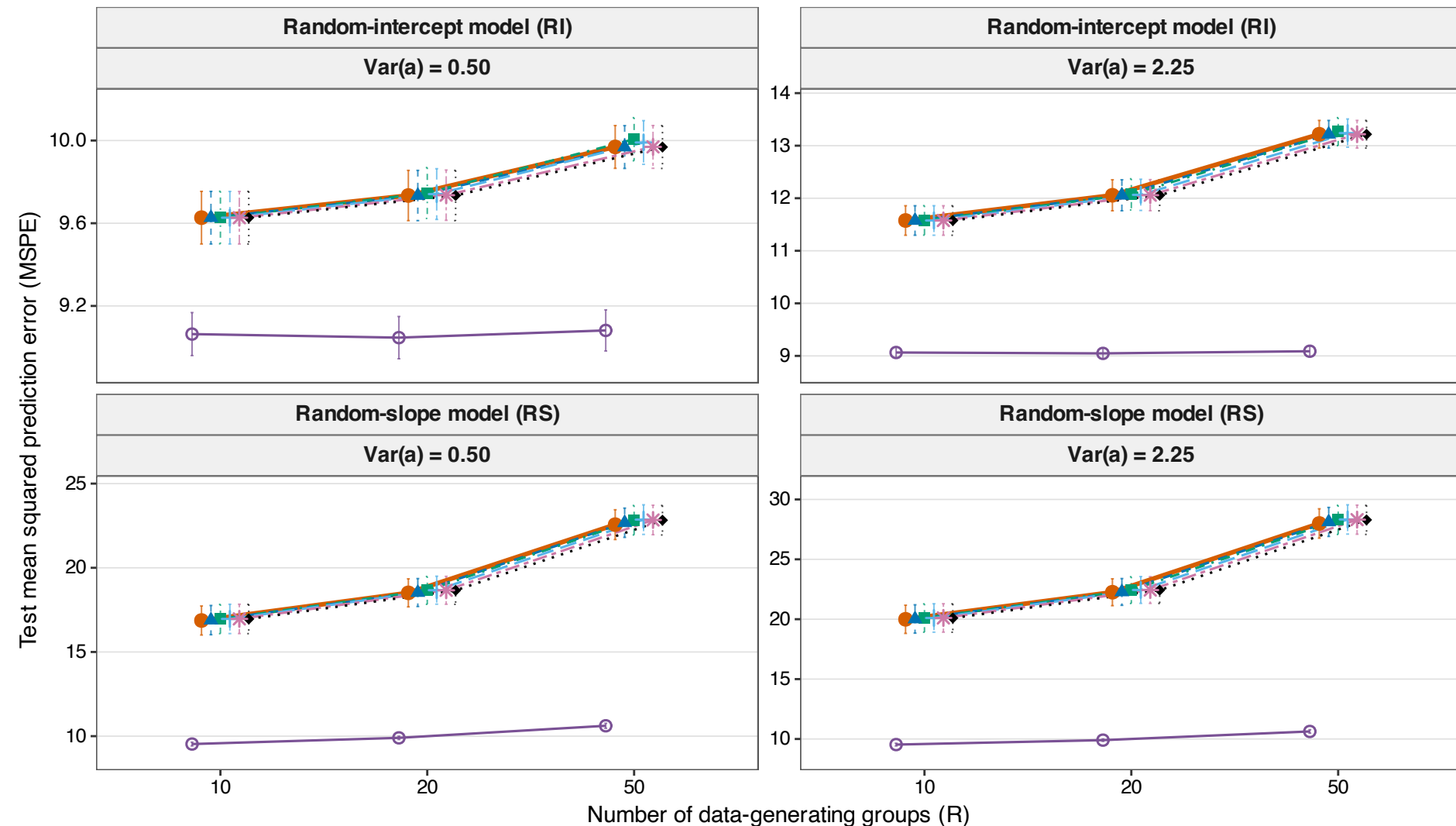
Mean test MSPE with 95% Monte Carlo error bars; each panel has an independent y-scale



Oracle uses the true generating groups and is an infeasible benchmark; all other methods use no true-group labels.

Case 15: Rare high-loss state

Mean test MSPE with 95% Monte Carlo error bars; each panel has an independent y-scale



ORACLE CIRG KM BLM CPF OBS LM

Oracle uses the true generating groups and is an infeasible benchmark; all other methods use no true-group labels.

Case 16: Target-distribution shift

Mean test MSPE with 95% Monte Carlo error bars; each panel has an independent y-scale

